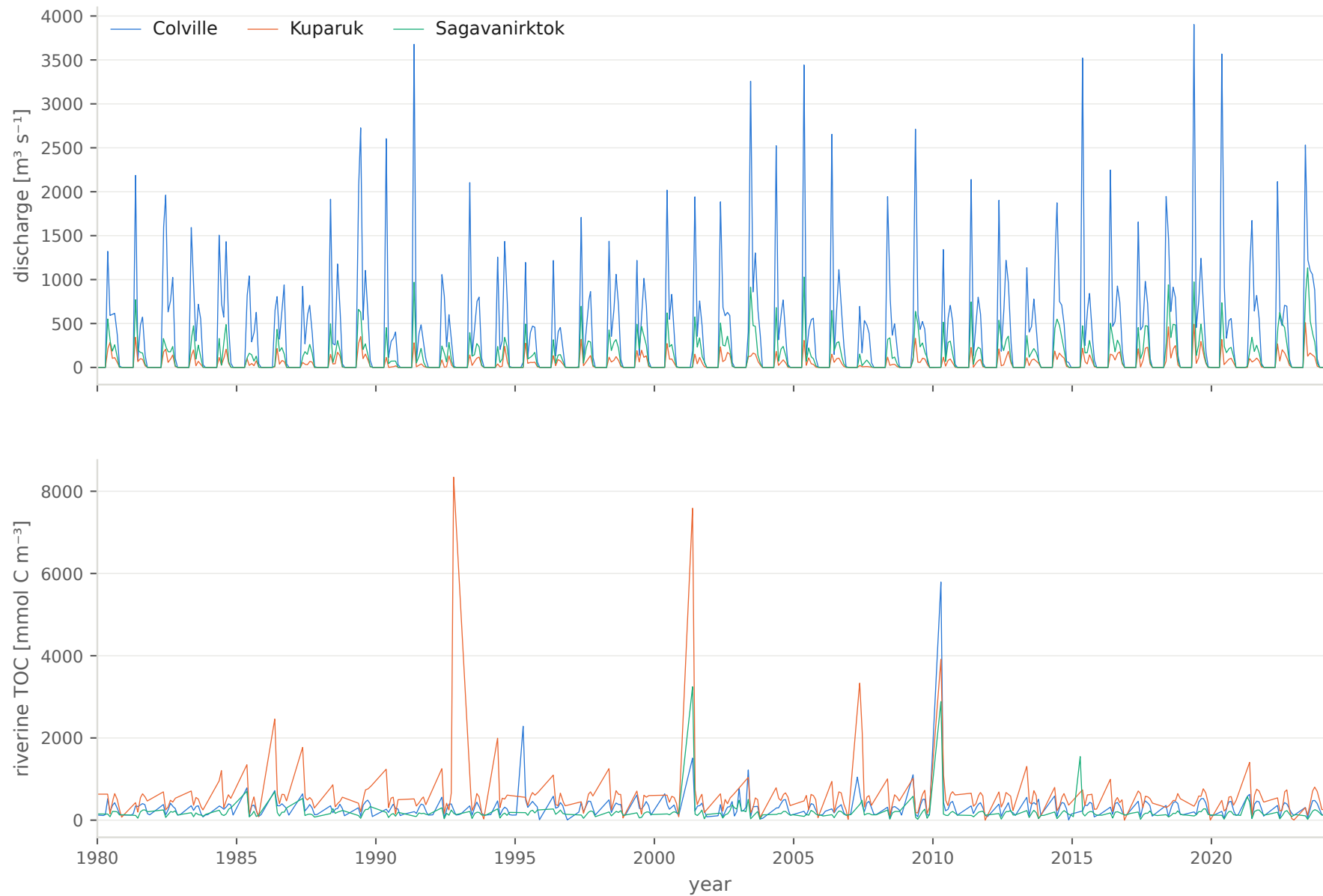


Interannual forcing, 1980-2023 (PWBM-derived)

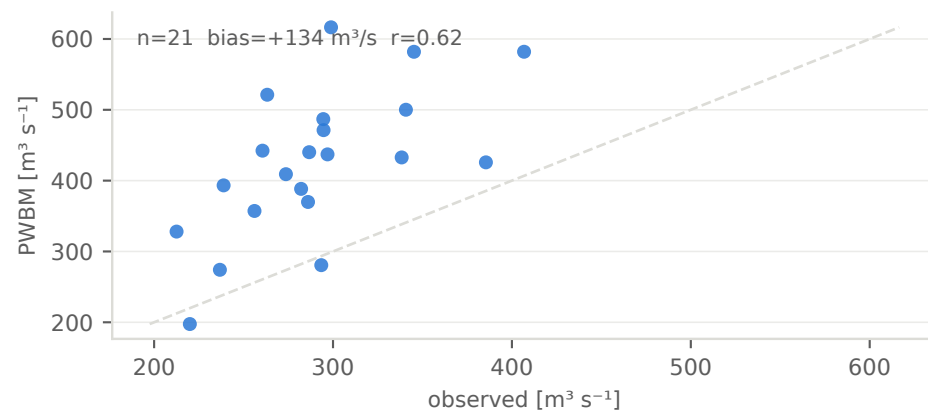
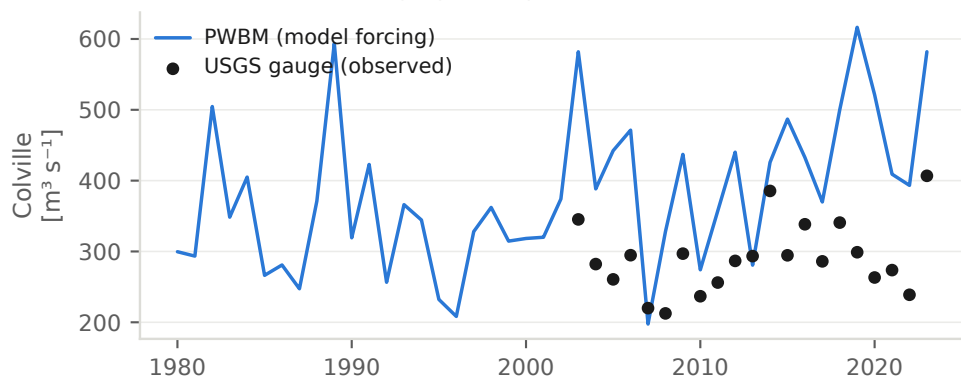
Discharge and riverine TOC (from PWBM DOC — see CLAUDE.md → "Interannual forcing") for the three runnable rivers. Everything else (met, tides, marine b



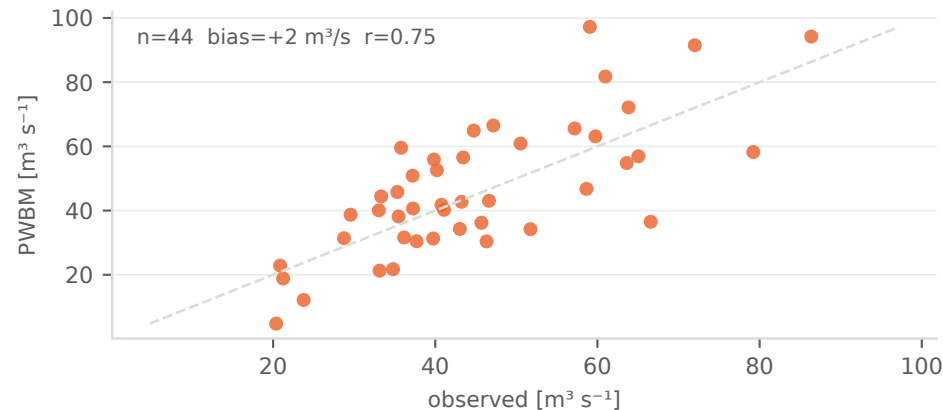
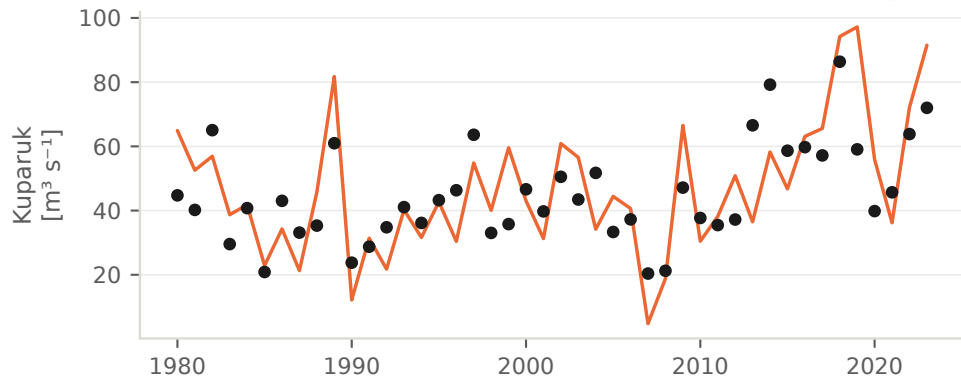
Discharge forcing vs. observation, 1980-2023

PWBM-derived discharge (the model's actual forcing, solid line) against the real USGS gauge annual mean (points), same three gauges tools/fetch_discharge.py uses

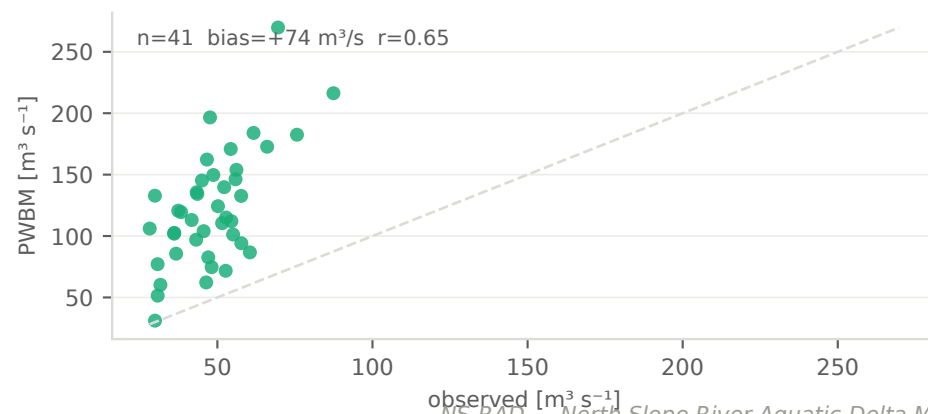
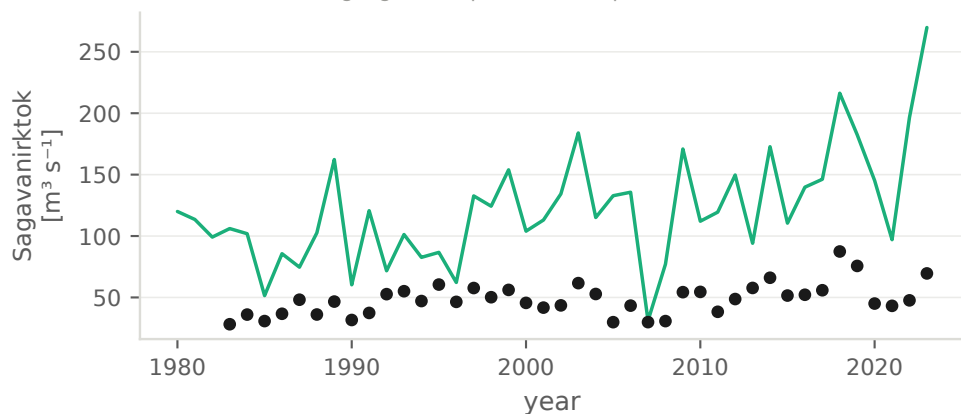
gauged far upstream (Umiat) — understates mouth flow



near tidewater — best-constrained comparison

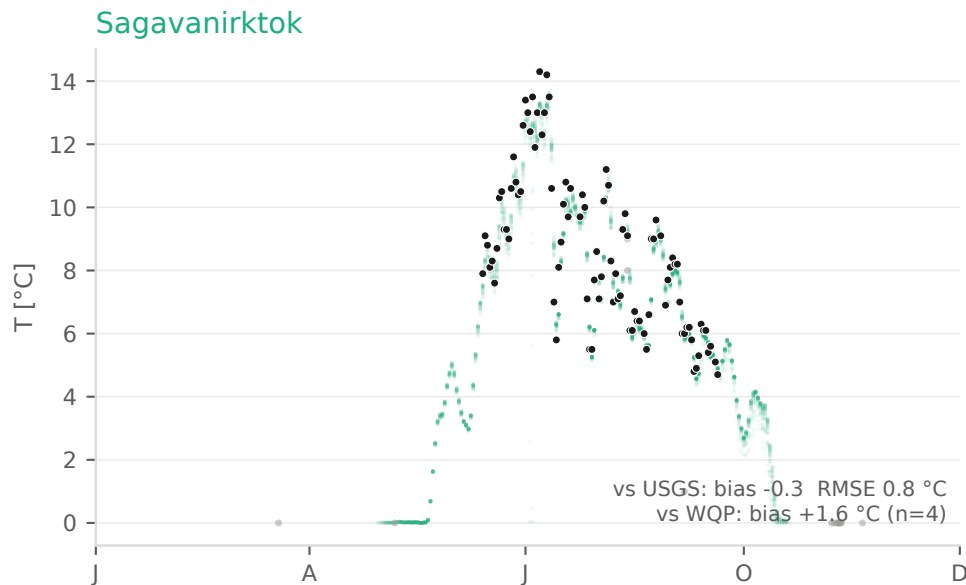
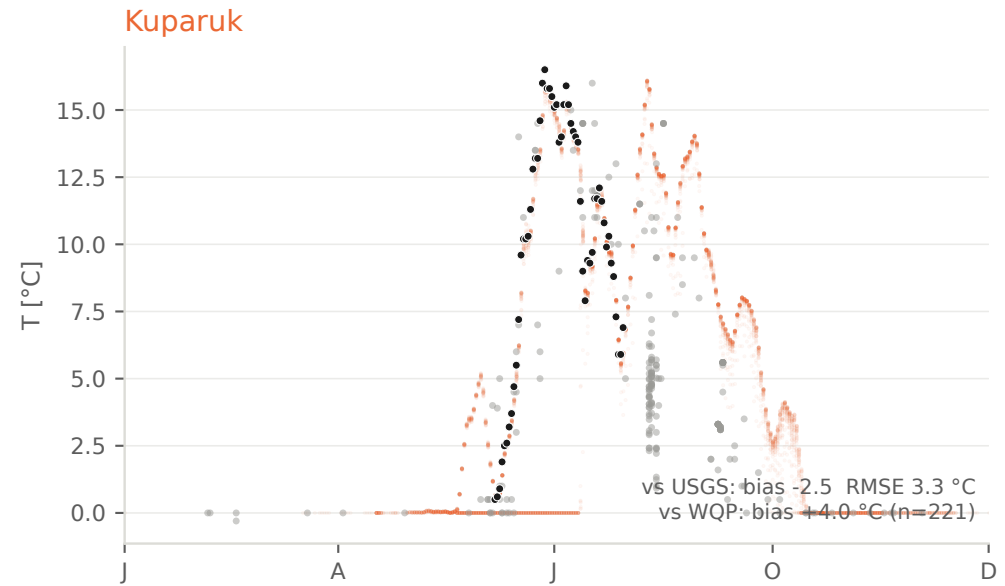
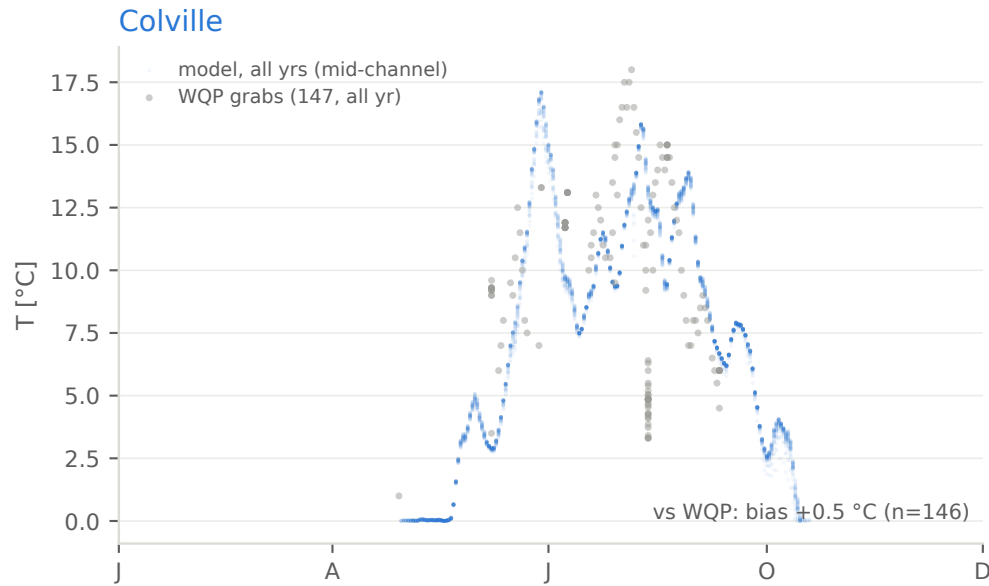


gauged far upstream (Pump Sta 3) — understates mouth flow

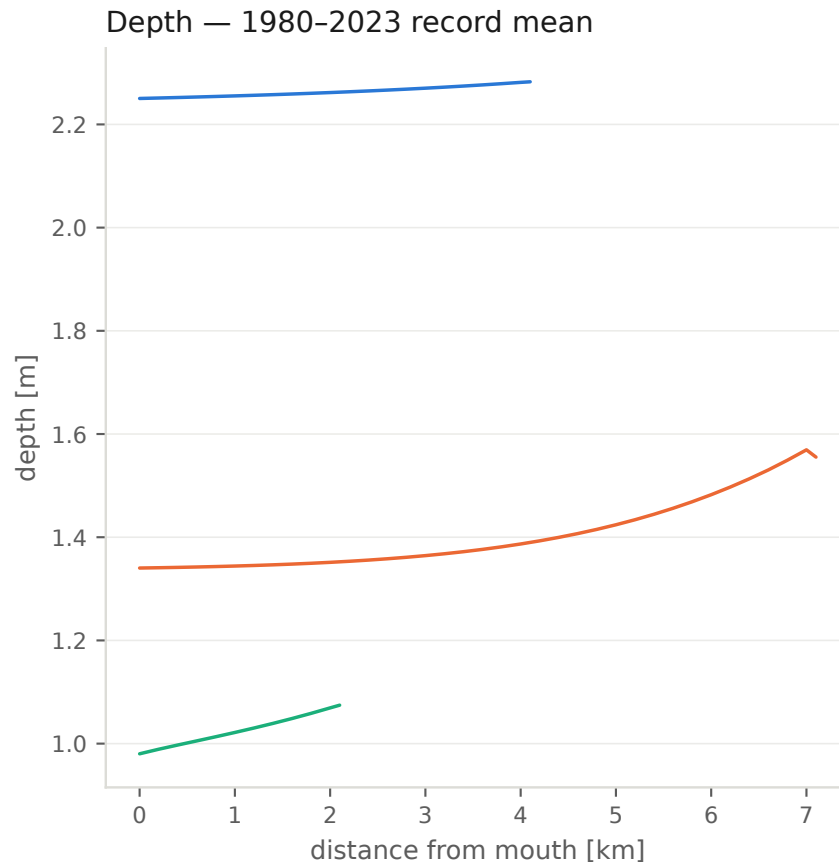
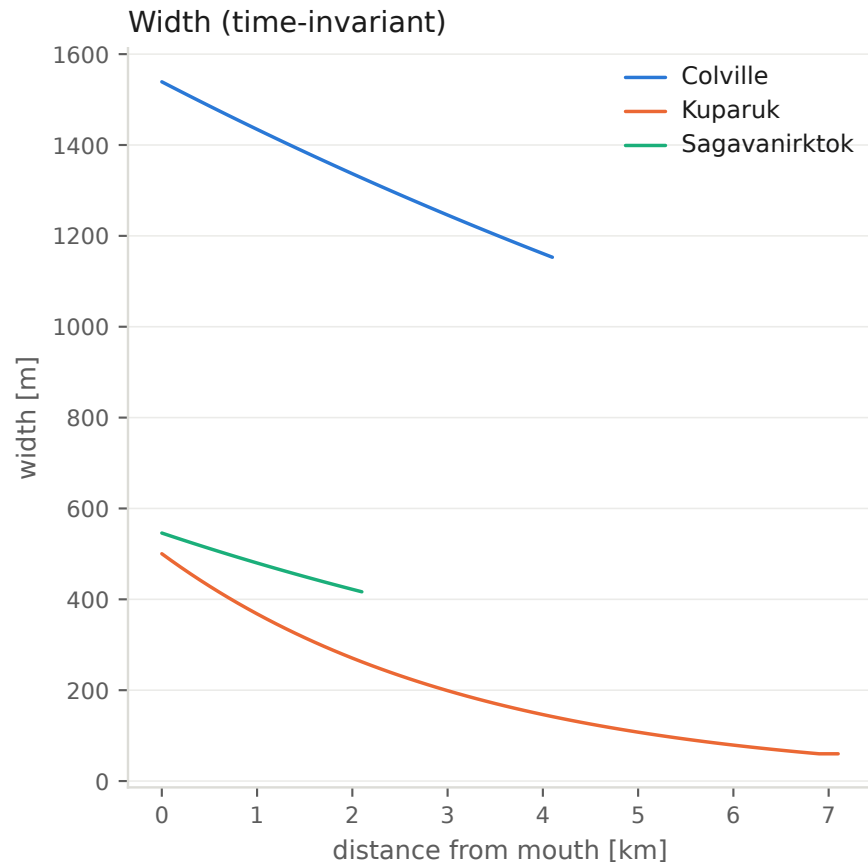


Water temperature — model (all 43 years) vs. observation

Every simulated day, mid-channel, folded to day-of-year and pooled across all 43 post-warmup years (semi-transparent — density shows where the interannual spread)

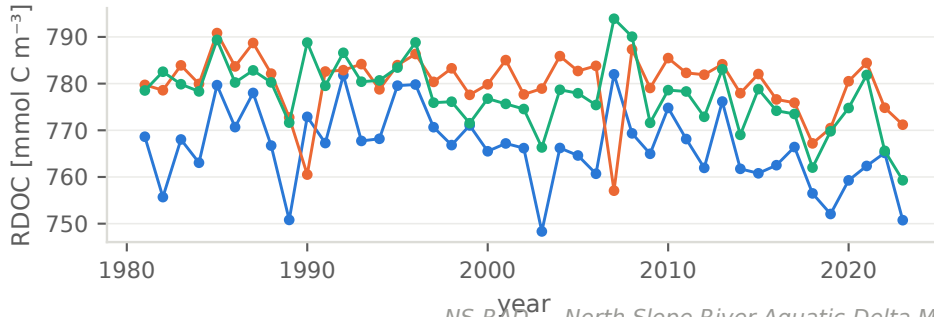
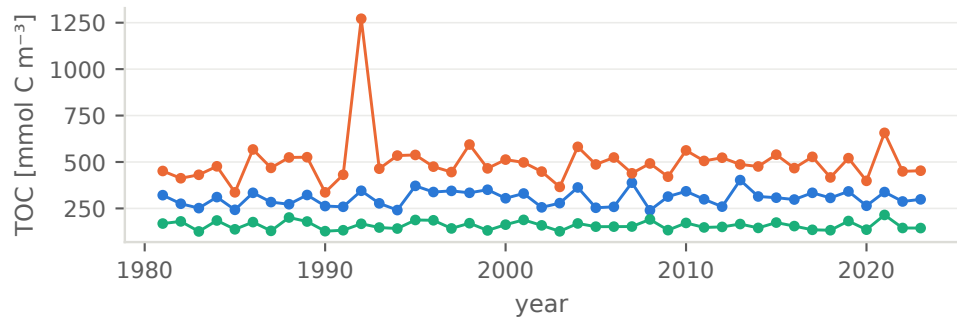
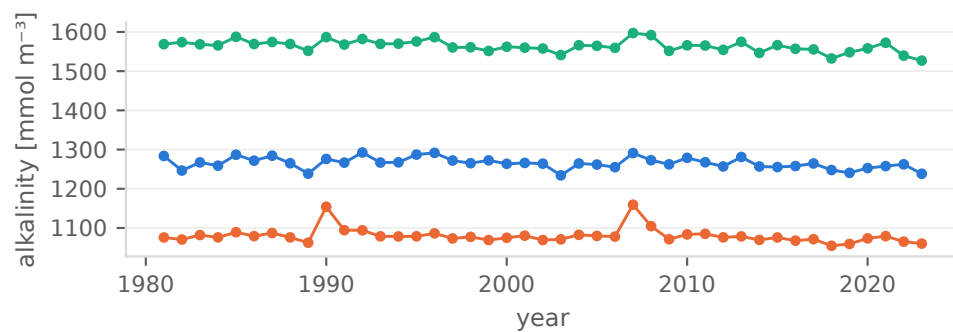
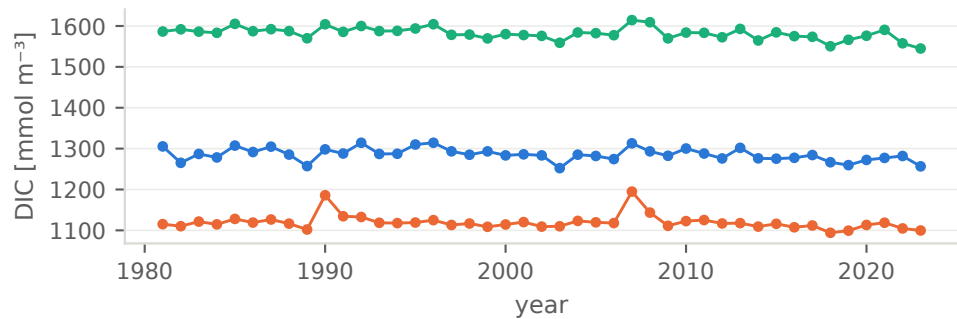
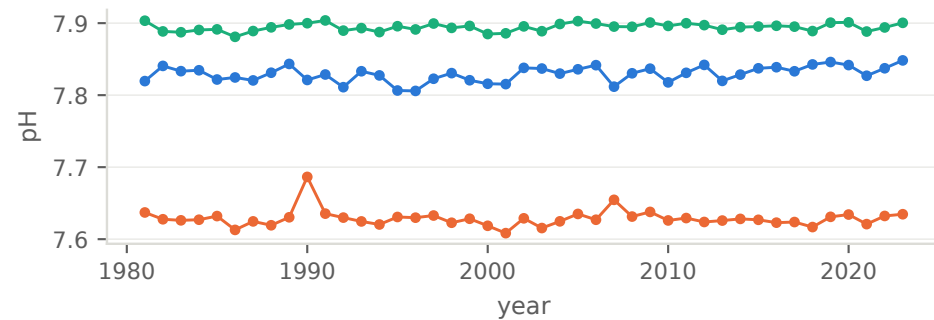
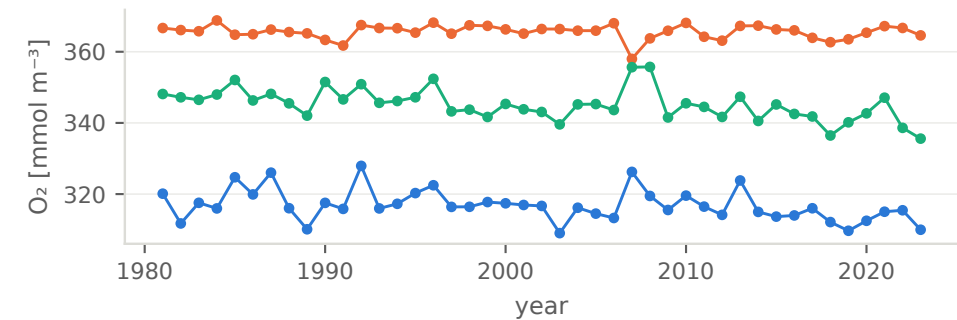
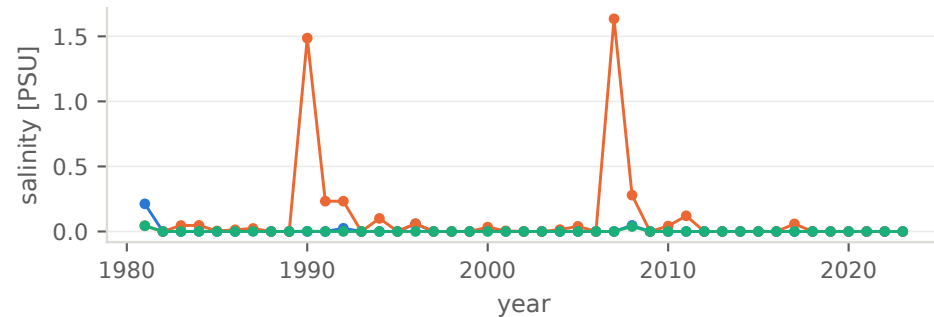
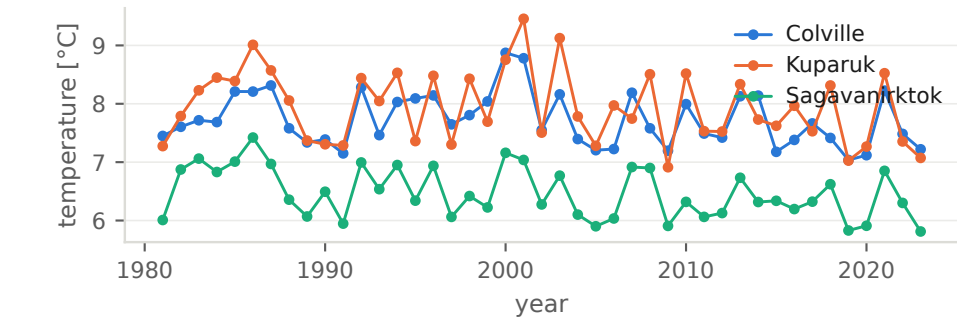


Channel geometry (reference — unchanged from the regular site)



Modelled state variables — annual open-water mean, 1981-2023 (1/2)

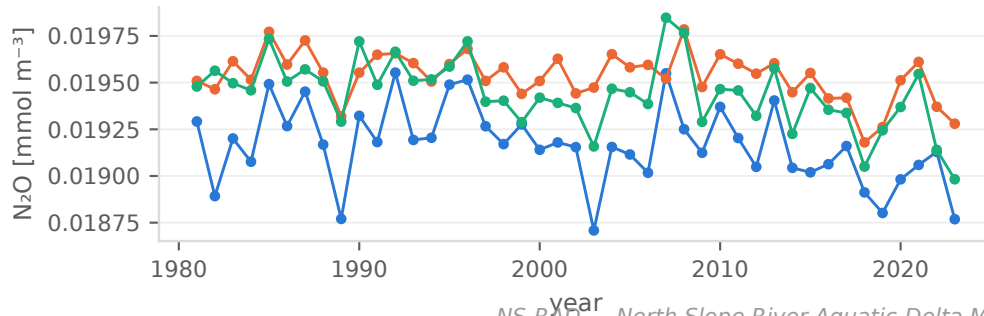
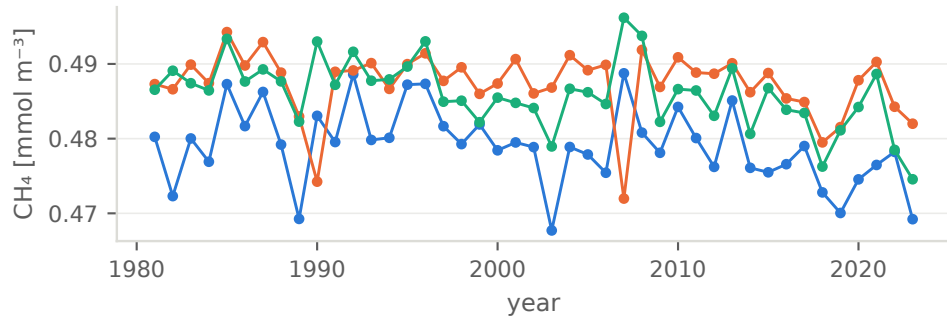
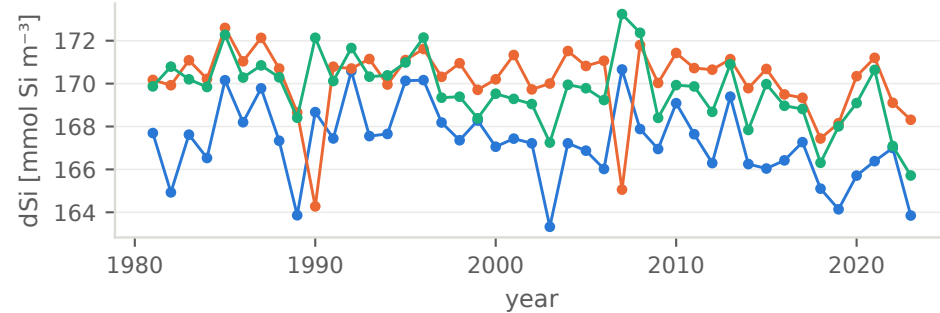
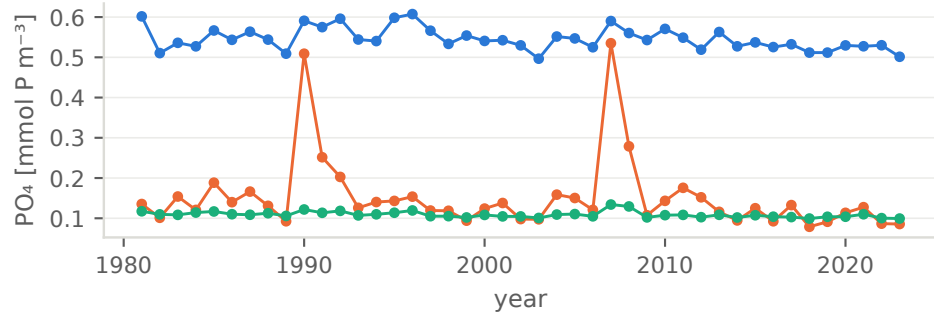
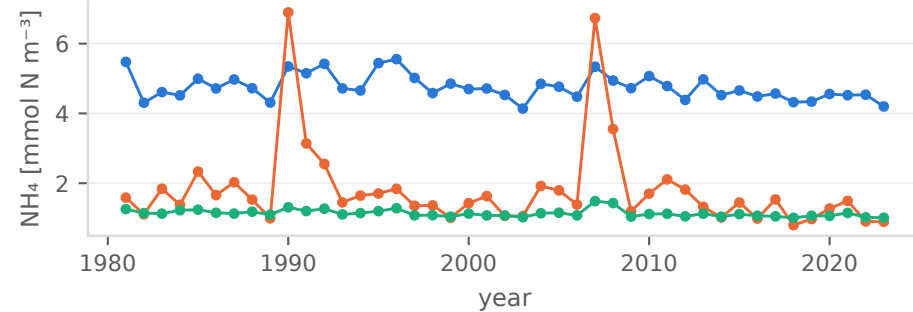
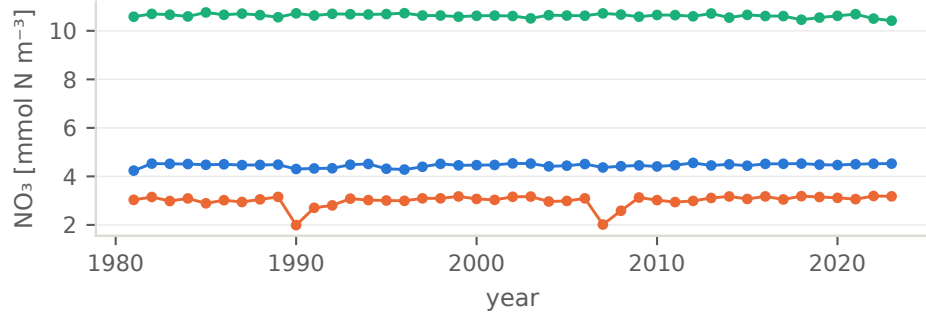
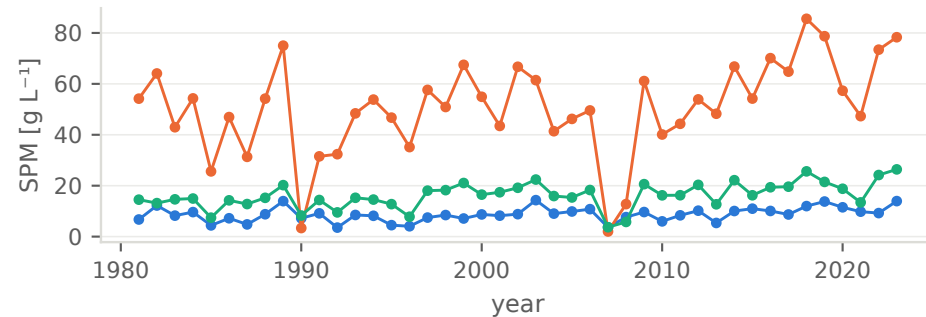
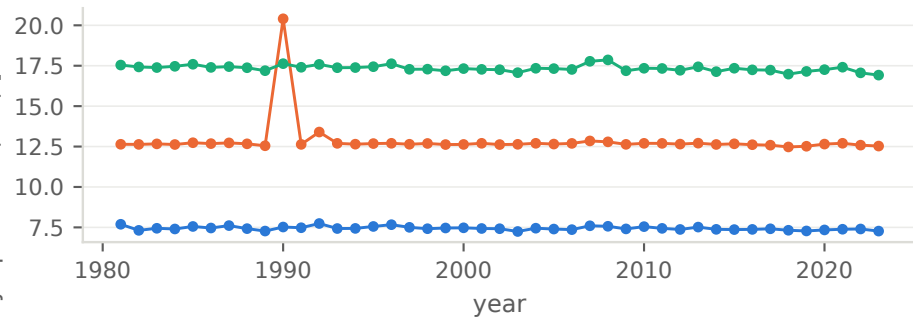
Every tracked species (config.py's species registry), spatial mean over open-water cells (ice_frac < 0.5), averaged across each full calendar year — how the



Modelled state variables — annual open-water mean, 1981-2023 (2/2)

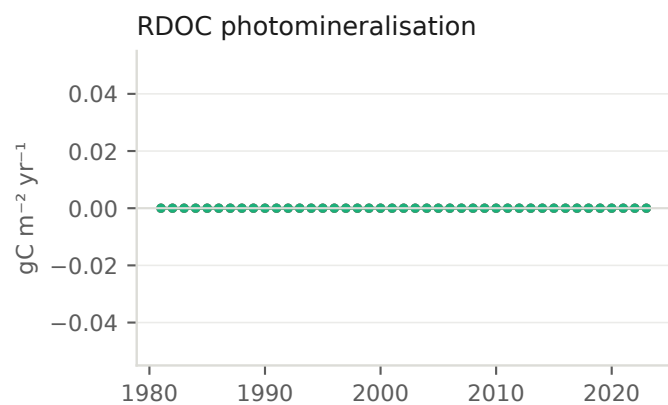
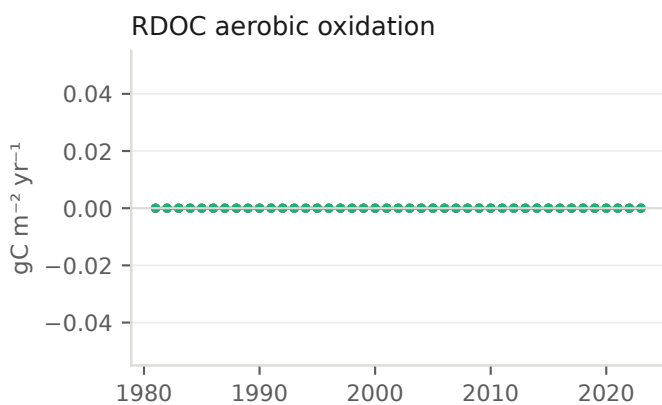
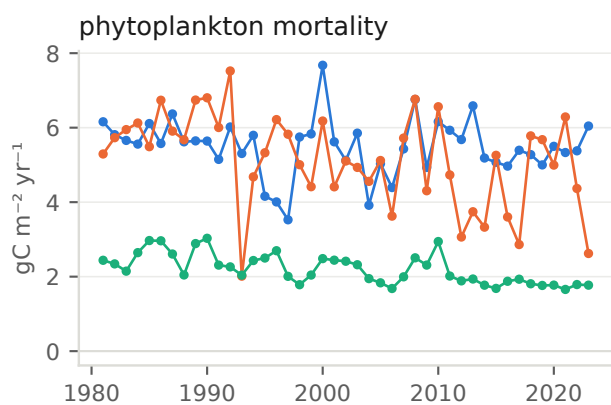
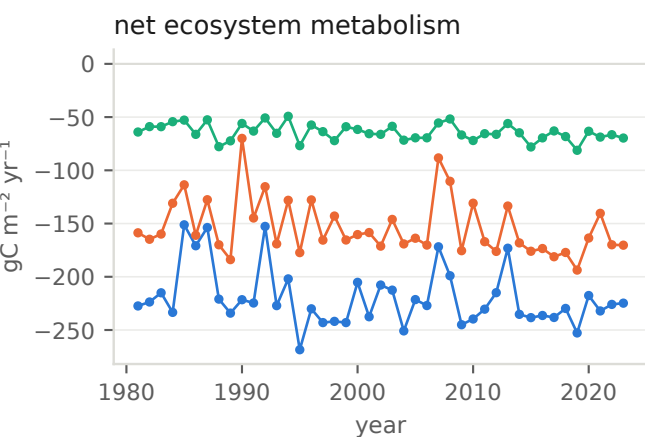
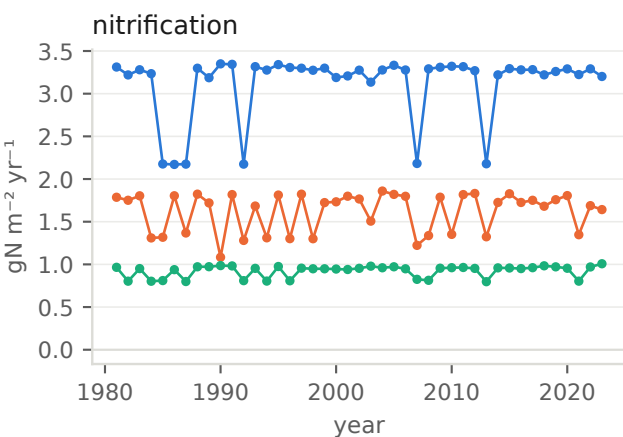
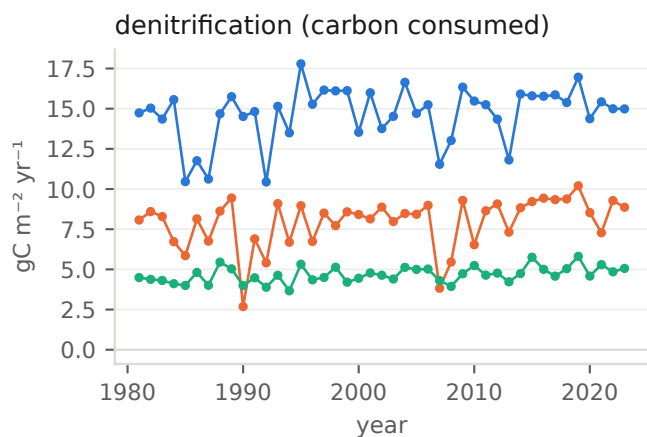
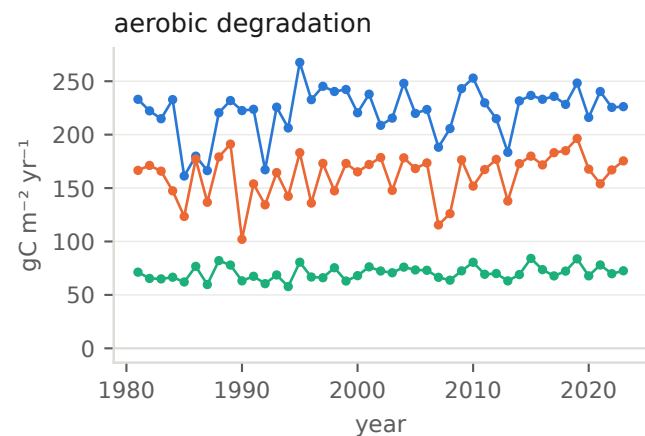
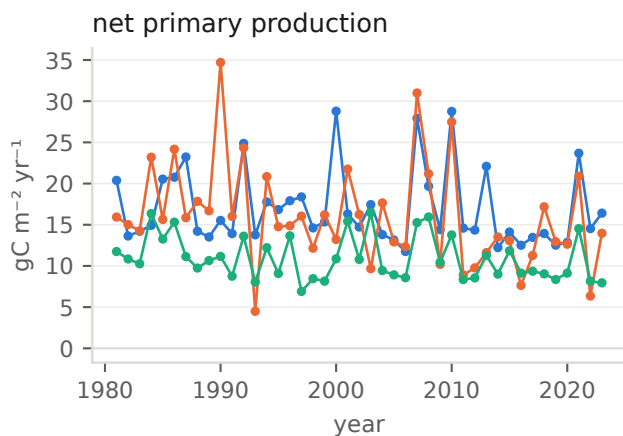
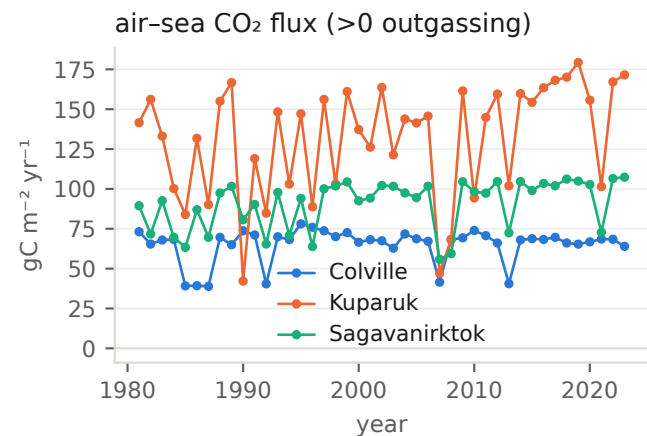
Every tracked species (config.py's species registry), spatial mean over open-water cells ($\text{ice_frac} < 0.5$), averaged across each full calendar year — how the

phytoplankton (DIA) [mmol C m^{-3}]



Model response: area-normalized annual budgets, 1981-2023 (1/2)

Every process rate written to output.nc. Same method as ns_rad_diagnostics.pdf's single-year area-normalized page (depth-integrate a volumetric rate, width



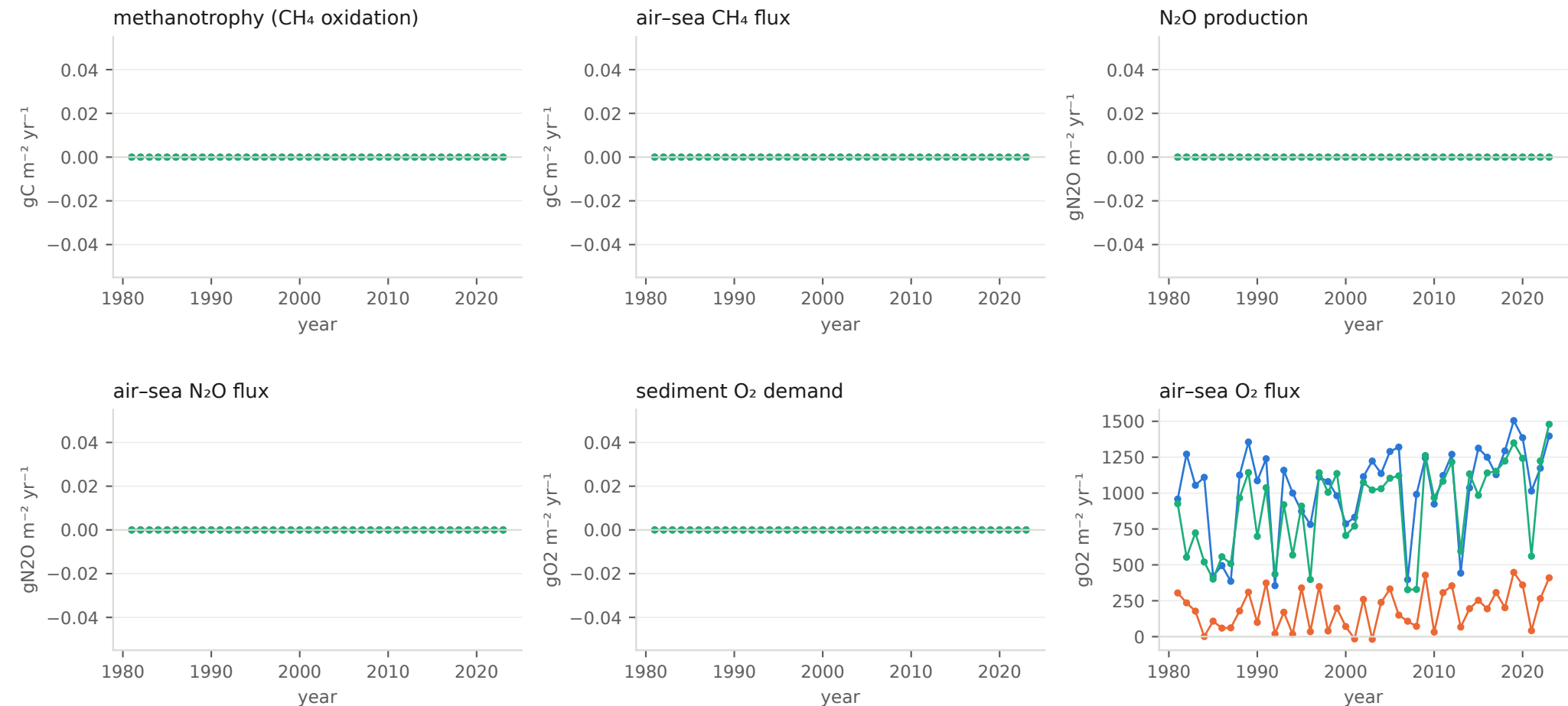
Surface area: Colville 5.481 km² · Kugaruk 1.445 km² · Sagavanirktok 1.005 km²

rdoc_ox / photo / ch4_ox / ch4_ex / n2o_prod / n2o_ex / sod are flat zero — config.ARCTIC_BGC is off (default) for these sites, so the Arctic extension reactions never run; RDOC/CH₄/N₂O still

NS-RAD — North Slope River-Aquatic-Delta Model

Model response: area-normalized annual budgets, 1981-2023 (2/2)

Every process rate written to output.nc. Same method as ns_rad_diagnostics.pdf's single-year area-normalized page (depth-integrate a volumetric rate, with



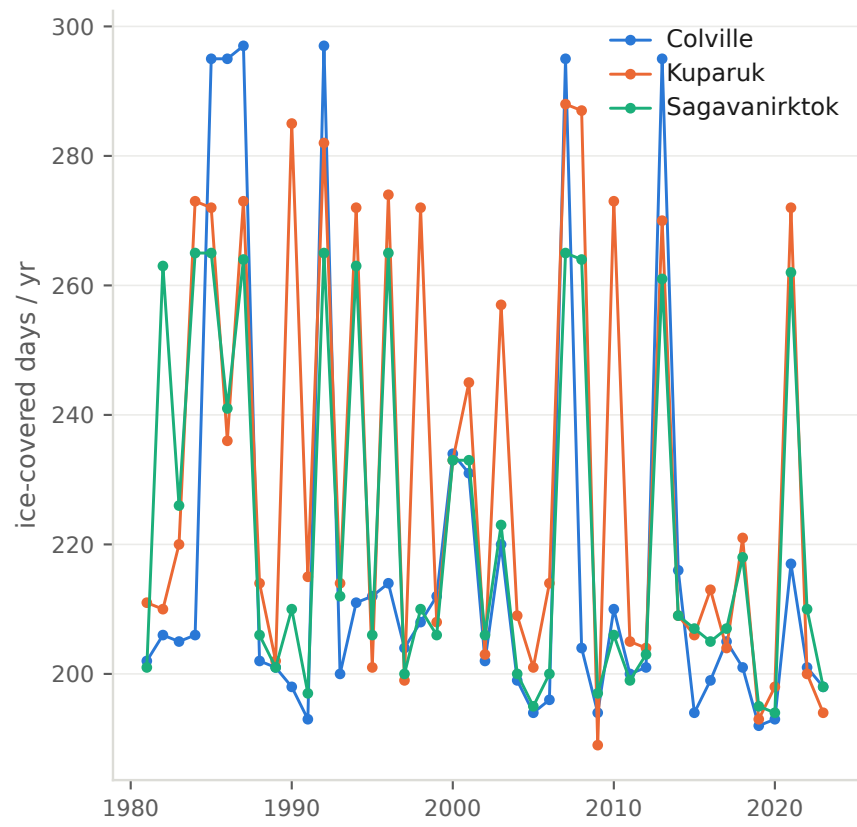
Surface area: Colville 5.481 km² · Kuparuk 1.445 km² · Sagavanirktok 1.005 km²

rdoc_ox / photo / ch4_ox / ch4_ex / n2o_prod / n2o_ex / sod are flat zero — config.ARCTIC_BGC is off (default) for these sites, so the Arctic extension reactions never run; RDOC/CH₄/N₂O still

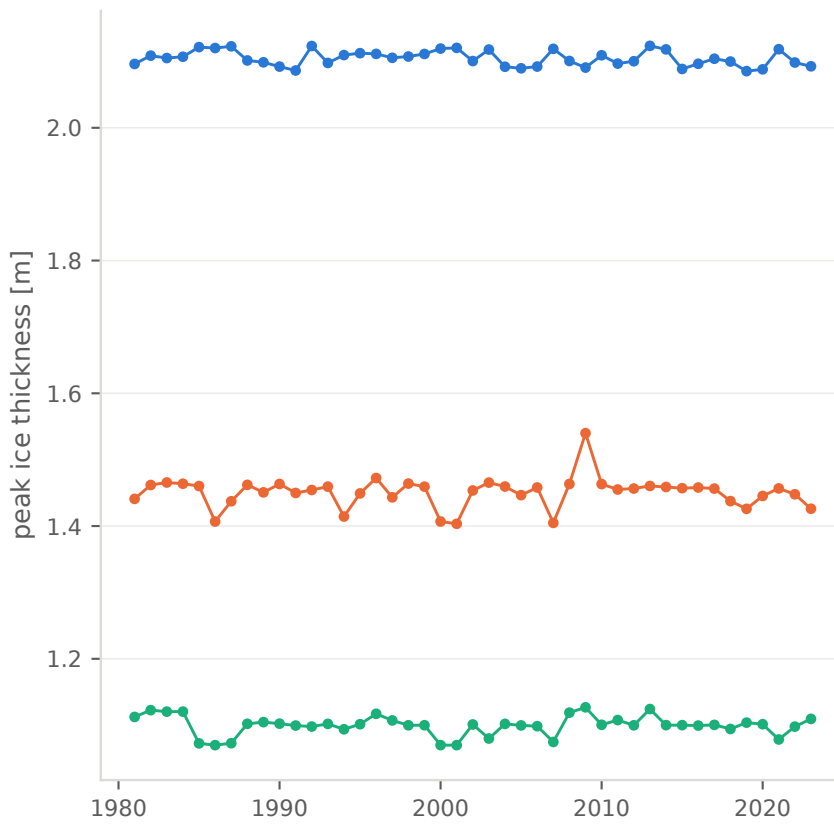
NS-RAD — North Slope River-Aquatic-Delta Model

Ice variability, 1981-2023

The hydraulic breakup mechanism (coefficient $BREAKUP_Q_FACTOR = 3.0 \times q_{ref}$) makes breakup timing a downstream consequence of the interannual discharge



Every year (all 3 rivers, 1981-2023) breaks up — the fixed-threshold concern above does not materialize in practice.



Does the model's response track the forcing?

